



Targeting sialic acid residues on lung cancer cells by inhalable boronic acid-decorated albumin nanocomposites for combined chemo/herbal therapy

Mayada M. Elgohary^{a,b}, Maged W. Helmy^{a,c}, Elsayeda-Zeinab A. Abdelfattah^d, Doaa M. Ragab^b, Sana M. Mortada^{a,b}, Jia-You Fang^{e,f,g,**}, Ahmed O. Elzoghby^{a,b,h,i,*}

^a Cancer Nanotechnology Research Laboratory (CNRL), Faculty of Pharmacy, Alexandria University, Alexandria 21521, Egypt

^b Department of Industrial Pharmacy, Faculty of Pharmacy, Alexandria University, Alexandria 21521, Egypt

^c Department of Pharmacology and Toxicology, Faculty of Pharmacy, Damanhour University, Damanhour, Egypt

^d Medical Research Institute (MRI), Alexandria University, Alexandria 21527, Egypt

^e Pharmaceutics Laboratory, Graduate Institute of Natural Products, Chang Gung University, Taoyuan 333, Taiwan

^f Research Center for Industry of Human Ecology and Research Center for Chinese Herbal Medicine, Chang Gung University of Science and Technology, Kweishan, Taoyuan 333, Taiwan

^g Department of Anesthesiology, Chang Gung Memorial Hospital, Kweishan, Taoyuan 333, Taiwan

^h Division of Engineering in Medicine, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115, USA

ⁱ Harvard-MIT Division of Health Sciences and Technology, Cambridge, MA 02139, USA

ARTICLE INFO

Keywords:

Albumin nanoparticles
Inhalable nanocomposites
Drug-phospholipid complex
Phenylboronic targeting
Pulmonary drug delivery
Lung cancer

ABSTRACT

Etoposide (ETP), as a potential treatment for lung cancer, has limited application due to its poor solubility, and systemic side effects. In the current study, we propose inhalable boronate-targeted HSA nanocomposites for combined delivery of ETP and the herbal drug, berberine (BER) for localized therapy of lung cancer. First, ETP was pre-formulated as phospholipid complex (EPC) to enhance drug solubility and facilitate its encapsulation within the hydrophilic albumin nanoparticles (NPs). Second, EPC and BER were then co-loaded with high efficiency into HSA NPs as a synergistic therapy for lung cancer. The NPs displayed suitable size around 200 nm and sequential drug release pattern. Moreover, conjugation of aminophenylboronic acid (APBA) to HSA NPs resulted in enhanced cytotoxicity and internalization into A549 lung cancer cells, compared to non-targeted NPs or free drugs via binding to sialic acid residues over-expressed by cancer cells. Using mannitol as a spray-drying carrier, the developed inhalable nanocomposites demonstrated deep pulmonary deposition, confirmed by small MMAD (2.112 μm) and high FPF (77.86%). In vivo investigations in lung cancer animal models revealed the superior anti-tumor efficacy of the inhalable nanocomposites. Overall, the inhalable APBA-HSA nanocomposites offered an alternative strategy for systemic delivery of ETP and BER in lung cancer therapy.

1. Introduction

Lung cancer is considered to be the most serious type of cancer, as it presents the highest percentage of cancer-related death worldwide. Conventional chemotherapy and radiation are widely used as a protocol of treatment. However, those strategies suffer from non-specificity to cancer cells and serious side effects which limit their application. Therefore, considerable research is being directed towards the development of a novel regimen to overcome these limitations [1].

As a promising alternative to systemic therapy, inhalational chemotherapeutics provide a localized therapy for lung tumors with the

benefit of avoiding the side effects of intravenous administration and delivery of high drugs doses [2]. Nanoparticles (NPs) encapsulating anti-cancer drugs possess the advantages of high cellular penetration, avoiding uptake by macrophages, increased drug loading and sustained release behavior [3–5]. However, the small particle size of NPs leads to their quick exhalation thus limiting their application for pulmonary administration [6]. On another avenue, development of inhalable microparticles encapsulating cytotoxic compounds ensures efficient lung deposition [7]. This signifies the rationale of developing nanoparticles-in-microparticles (nanocomposites), as they possess the advantages of both microparticles, to enable effective aerosolization and deep lung

* Correspondence to: Ahmed O. Elzoghby, Division of Engineering in Medicine, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115, USA.

** Correspondence to: Jia-You Fang, Graduate Institute of Natural Products, Chang Gung University, Taoyuan 333, Taiwan.

E-mail addresses: fajy@mail.cgu.edu.tw (J.-Y. Fang), aelzoghby@bwh.harvard.edu, ahmed_elzoghby@alexu.edu.eg (A.O. Elzoghby).

<https://doi.org/10.1016/j.jconrel.2018.07.014>

Received 22 March 2018; Received in revised form 6 July 2018; Accepted 9 July 2018

Available online 25 July 2018

0168-3659/ © 2018 Elsevier B.V. All rights reserved.

deposition, and nanoparticles in targeting the tumor cells [8].

Etoposide (ETP) is a potent inhibitor of topoisomerase II used for treatment for non-small cell lung cancer (NSCLC) [9]. However, the dose-dependent side effects of ETP treatment include reduced count of white blood cells and platelets, and secondary leukemia [10]. Additionally, resistance to ETP could occur due to up-regulation or mutation of topoisomerase II molecule and the rapid repair of DNA breaks [11]. Cellular extrusion of ETP mediated by transport systems could lead to cross-resistance against other cytostatic agents, as observed for MDR-1-mediated multidrug resistance [12]. Clinical studies revealed that combinatorial cancer treatment significantly reduces individual drug-related toxic effect and evade multi-drug resistance (MDR) via different pathways [13]. In the present study, ETP was combined with the natural isoquinoline alkaloid berberine (BER) as a unique drug combination for effective treatment of lung cancer. BER was previously reported to modulate the MDR through inhibition of P-gp and/or MRP efflux activity in A549 lung cancer cells [14]. Moreover, BER was found to inhibit DNA topoisomerases and protein kinase C, sensitize cancer cells to drugs by increasing their ROS levels and reversal of MDR, thus elevates the cytotoxic effect of combined anticancer drugs [14, 15].

Among promising nanocarriers, human serum albumin (HSA) has attracted great attention in the preparation of NPs due to its biocompatibility, biodegradability, non-immunogenicity, and stability. Moreover, HSA can be chemically modified through its amine and carboxylate moieties for active targeting to cancer cells, beside its targeting abilities through transcytosis pathway by binding to gp60 and SPARC proteins [16, 17]. Unfortunately, hydrophobic anti-cancer drugs are not easily incorporated in hydrophilic albumin NPs except using high energy and shearing technology such as high pressure homogenization (HPH) in case of nab-paclitaxel (Abraxane®) approved by FDA for treatment of advanced and metastatic NSCLC [18]. Therefore, novel pre-formulation approaches may be required to enhance the solubility of hydrophobic drugs and hence facilitate their loading into albumin NPs.

Among those solubility enhancing approaches, complexes between drugs and native phospholipids like soybean phosphatidylcholines, are formed based on hydrogen bonding between the polar parts in both phospholipids and drug [19–21]. The water-insoluble cytotoxic drug, teniposide was pre-formulated as phospholipid complex prior to incorporation into albumin NPs [22]. In addition to the poor water solubility, another challenge lies in non-selective biodistribution of chemotherapeutics leading to systemic toxicity. Therefore, active-targeted nanocarriers were designed to enhance drug accumulation into tumor cells [23, 24]. The presence of over-expressed sialic acid on cancer cell surface is implicated in metastasis, progression, and resistance to chemotherapy [25]. Those sialic acid residues provide a potential binding site for boronic acid group-modified nanocarriers leading to enhanced intra-tumoral transport of nanocarriers through sialic acid recognition [26].

In comparison to the previously reported ETP-albumin NPs prepared by high pressure homogenization (HPH), the desolvation approach does not produce an increase in temperature that may affect the stability of incorporated drugs or induce changes in the protein structure with disruption of some disulfide bonds, thus varying the availability of hydrophobic groups [27, 28]. Moreover, NPs prepared by HPH suffer from low stability and high probability of dis-assembly in the systemic circulation [29]. Concerning injectable APBA-targeted NPs, normal cells may express traces of sialic acid residues for cellular/extracellular interaction which makes the NPs not entirely selective to cancer cells [30]. In addition, higher i.v. doses should be administered compared to those delivered locally via inhalation, thus raising the systemic toxicity and increasing the drug costs. Although nebulized NPs have been widely used for pulmonary administration, they suffer from suspension instability and strong particle-particle interactions, leading to non-adequate nebulization and exhalation of low-inertia NPs [31]. Inhalable NPs powder could solve the stability issue but still suffers

from non-suitability for deep lung delivery [6]. Our inhalable nanocomposites offered optimum size for deep lung deposition compared to previously reported inhalable ETP delivery systems [2, 9]. In addition, the NPs would be recovered from microparticles only after reaching deep lung tissues and selectively target cancerous cells offering maximal antitumor efficacy.

So far, the current study is the first to develop inhalable dry powder composed of spray-dried APBA targeted-albumin nanocomposites for combined delivery of ETP and BER to lung cancer. First, to facilitate its incorporation into the hydrophilic HSA matrix without solubilizer, the hydrophobic chemotherapeutic drug, ETP was pre-formulated as phospholipid complex (EPC) by direct lyophilization. Second, for enhancing ETP anti-tumor efficacy, and reducing its therapeutic dose thus minimizing its toxicity and prevent the development of resistance, the water-soluble herbal drug, BER was directly co-encapsulated with EPC into HSA NPs via desolvation method using different crosslinkers. Third, maximal targeting efficacy was hypothesized by sialic acid targeting via anchoring phenylboronic acid to the surface of HSA NPs. Finally, to reduce systemic toxicity, inhalable dry powder nanocomposites were developed via co-spray drying of dual-drug-loaded HSA NPs with inert carriers to prepare micro-sized powders suitable for deep lung deposition. The developed delivery system was thoroughly investigated *in vitro* and *in vivo* to prove the anti-tumor superiority of the combined drug nano-delivery compared with the free drug combination.

2. Materials and methods

2.1. Materials

Human serum albumin (HSA), Berberine chloride hydrate (BER), 3-Aminophenyl boronic acid (APBA), tertiary Butyl Alcohol (TBA), N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDC), N-Hydroxysuccinamide (NHS), sodium hydroxide (NaOH), glutaraldehyde solution (GA, 25%), zinc nitrate hexahydrate (Zn (NO₃)₂·6H₂O), Sodium Chloride (NaCl), mannitol, maltodextrin, trehalose, L-leucine, 3-(4,5-dimethylthiazolyl-2)-2,5-diphenyltetrazolium bromide (MTT), Fluorescein isothiocyanate (FITC), urethane (ethyl carbamate), Haematoxylin solution, Eosin solution (H&E) and Canada balsam were purchased from Sigma-Aldrich (St. Louis, USA). Etoposide (ETP) and genipin (GNP) were purchased from Xian Natural Field Bio-Technique Co., Ltd. (Xian, China). Fat-free soybean phospholipids with 70% phosphatidylcholine (Lipoid S75) were kindly provided by Lipoid GmbH (Ludwigshafen, Germany). Poly (ethylene glycol) 400 (PEG-400) was supplied by Pharonia Pharmaceuticals (Alexandria, Egypt). Absolute ethanol and methanol were purchased from ADWIC, El-Nasr Pharmaceutical Chemicals Co. (Cairo, Egypt). Methanol and acetonitrile HPLC grade were purchased from JT Baker (Phillipsburg, NJ, USA).

2.2. Preparation and characterization of EPC

ETP-PC complex (EPC) was prepared by direct freeze-drying method [32]. Different molar ratios (1:1, 1:2 and 1:4) of ETP and phospholipid (Lipoid S75) were dissolved in tertiary butyl alcohol (TBA, 10 ml). The mixture was maintained at room temperature on a magnetic stirrer until complex formation. The complex was frozen at −80 °C for 4 h then lyophilized for 24 h using mannitol as a cryoprotectant (LyoQuest-55 freeze-dryer, Telstar, Spain). The prepared EPC was evaluated for their physicochemical, and solid state characteristics [33] (Detailed in the Supporting Information). To confirm the enhanced solubility of EPC, the dissolution profiles of EPC prepared with different molar ratios of phospholipid were compared to free ETP. The optimized EPC powder (P1 and P4), equivalent to 5 mg ETP, was added to 50 ml PBS (pH 7.4) and shaken at 100 rpm and 37 °C (Shaking water bath, Maxturdy 30, daihan scientific Co. Ltd., Shanghai, China). Fixed volume samples were

withdrawn at different time intervals, replaced with fresh medium for compensation and analyzed using HPLC.

2.3. Preparation of ETP/BER-loaded HSA NPs

EPC or EPC/BER-loaded HSA NPs were prepared via desolvation method [34, 35]. Briefly, HSA (200 mg) was dissolved in NaCl (2.0 ml, 10 mM) solution containing BER/EPC (20 and 30 mg, respectively) incubated for 1 h at room temperature. The solution pH was adjusted to 8.0 by 0.1 N NaOH. Drop-wise addition of absolute ethanol (8 ml) upon stirring (500 rpm) at a rate of 1 ml/min leads to NPs formation. Different crosslinker types were used to harden the NPs upon overnight stirring at room temperature. Purification of the produced NPs was performed by differential centrifugation (3–30 KS, Sigma Laborzentrifugen GmbH, Germany) at 16,500 rpm, 30 min, and 4 °C. The obtained NPs were re-suspended in distilled water using an ultrasonic bath (Daihan scientific Co. Ltd., Shanghai, China).

2.4. Preparation of APBA-targeted HSA NPs

APBA-targeted HSA NPs were prepared using APBA-HSA by the same procedures mentioned above. APBA was conjugated to HSA via EDC/NHS reaction to form amide bond Method [36]. The conjugation was confirmed by MALDI-TOF analysis, tryptophan fluorescence quenching, Alizarin Red S (ARS) test, FTIR, and DSC.

2.4.1. MALDI-TOF MS analysis

The conjugation of APBA to HSA was confirmed by MALDI-TOF MS analysis using Bruker Daltonics ultraflex TOF/TOF instrument (Bremen, Germany). The matrix system used was DHB (20 mg/ml) dissolved in TFA (0.1%) and acetonitrile mixture (70/30 v/v). Spectra were obtained in positive ion mode using an accelerating voltage of 25 kV and data analysis was performed using flexAnalysis software.

2.4.2. Tryptophan fluorescence spectra

Protein intrinsic fluorescence was obtained by tryptophan excitation at 295 nm and emission spectra were collected from 320 to 420 nm with 5 nm excitation and emission slits, using LS 55 fluorescence spectrometer (Perkin Elmer, USA). Each sample (10 mg) was dissolved in 50 mM PBS pH 7.2 and the absorbance of the samples was adjusted to 0.05 at 280 nm. Emission spectra of PBS were subtracted from those containing the samples and the conjugate emission spectra were normalized to native HSA.

2.4.3. Alizarin red test

An aqueous stock solution of 4×10^{-4} M ARS in phosphate buffer (pH 7.4) was prepared. 1.5 ml APBA-HSA (10 mg/ml) was added to 2.0 ml of ARS stock solution for 30 min. The change in color and absorbance was observed by UV/VIS spectroscopy.

2.5. Physicochemical characterization of HSA NPs

Particle size, surface charge, in vitro drug release and morphology of the prepared NPs were evaluated [4, 37] and detailed in the Supplementary section.

2.6. In vitro cytotoxicity and cellular uptake

By MTT assay, the in vitro cytotoxicity of free drugs solution, non-targeted and APBA-targeted ETP/BER-dual loaded HSA NPs were evaluated against A549 lung cancer cells. For localization of the NPs in cells, NPs were labeled with FITC according to a previously described method [38]. The FITC-labeled HSA NPs were purified from unreacted FITC by dialysis (12–14 kDa MWCO VISKING dialysis tubing, SERVA, Germany) for 48 h. Cellular uptake of free FITC, FITC-HSA NPs, and FITC-APBA-HSA NPs was investigated using confocal microscopy [39] (Supplementary section).

2.7. Preparation of inhalable spray-dried HSA nanocomposites

Drug-loaded HSA NPs were microencapsulated within different carrier matrices via spray-drying technique. The sedimented HSA NPs were re-dispersed in an aqueous solution containing the selected carrier combination and spray-dried using a laboratory scale B-290 Büchi spray-dryer (Flawil, Switzerland), equipped with a high-performance cyclone, two component nozzles, and co-current flow. For optimization of the process, the inlet air temperature (110 °C), the pump flow rate (10%), the aspirator (100%) and the atomizing air flow rate (320 l/h) were adjusted. Experimental variables including total solid concentration, carrier type, and HSA NPs/carrier ratio have been investigated.

2.8. Characterizations of the spray-dried nanocomposites

Yield, aqueous redispersibility, morphology, particle size, density, calculated aerodynamic properties, solid state characteristics, drug release and long-term stability were examined according to the methodology presented in Supplementary Section.

2.9. In vitro aerosol deposition of nanocomposites

Analysis of deposition profile of spray-dried HSA nanocomposites was performed using the eight stages Andersen Cascade Impactor (ACI; Copley Scientific Ltd., Nottingham, UK). ACI was operated at air flow rate of 28.5 l/min, with 4 kPa pressure drop. Budelizer (European Egyptian Pharm. Ind. Co., Alexandria, Egypt), a single-dose inhaler device, was connected to the ACI and filled with a hard gelatin capsule (size 2). Powder was actuated and withdrawn from capsules by the 28.5 l/min flow rate to be deposited in the device, throat, and each stage of ACI collection plates. Each actuation was set in 4 s and 2 aerosolization experiments were performed. HPLC mobile phase (10 ml) was used to wash the powders from the device and stages to be analyzed by HPLC. Aerodynamic parameters (FPF, MMAD, and GSD) were automatically calculated using Copley Inhaler Testing Data Analysis Software (CITDAS) (Supplementary Section).

2.10. In vivo study

2.10.1. Animals

For determination of the deposition and treatment efficiency of HSA nanocomposites, male Albino mice (15–20 g) were used following the animal care and use protocol of Alexandria University (Supplementary section).

2.10.2. In vivo lung deposition study

In order to investigate the in vivo lung deposition profile, the dry powder insufflator (DP-4 M, Penn Century Inc. PA, USA) was used for administration of the tested powders. Mice were anesthetized with intra-peritoneal (I.P.) injection of ketamine (12.5 mg/kg) and xylazine (1.5 mg/kg) and restrained in a supine position. FITC-APBA-HSA nanocomposite powder and free FITC were quickly administered via intratracheal route. After sacrifice of mice at different time points (4, 24, 48, and 72 h) post-administration, lungs were collected and washed by PBS. The tissue sections of lungs were imaged using fluorescence microscopy (Olympus BX41, Japan; excitation and emission λ_{max} at 495 and 520 nm, respectively). The concentration of the fluorescent nanocomposites relative to the fluorescence intensity in lung sections was analyzed by Image J software (Systat, San Jose, CA, USA).

2.10.3. Induction of lung cancer

Lung carcinoma was chemically-induced in 12–16 weeks male Albino mice weighing 15–20 g by receiving I.P. injection of urethane (1.5 g/kg), dissolved in 0.9% NaCl, followed by a booster dose after 8th week [40].

Table 1
Physicochemical properties of different ETP-phospholipid complex (EPC).^a

	ETP/lipid molar ratio	Mannitol (% w/v)	Lyophilization yield (%)	EE (%)	DL (%)	Particle size (nm)	PDI	Zeta potential (mV)
P1	1:1	0.5	76.2 ± 4.20	100.0 ± 0.08	21.8 ± 0.09	121.0 ± 2.70	0.208 ± 0.08	−27.9 ± 0.51
P2	1:2	0.5	82.2 ± 3.90	81.4 ± 2.35	15.0 ± 0.14	139.0 ± 5.20	0.593 ± 0.10	−42.0 ± 2.63
P3	1:4	0.5	82.4 ± 3.30	91.2 ± 3.60	11.4 ± 0.19	251.0 ± 8.06	0.420 ± 0.10	−43.2 ± 1.98
P4	1:1	1.0	95.0 ± 2.98	95.3 ± 3.10	12.6 ± 0.10	140.0 ± 3.79	0.200 ± 0.07	−27.0 ± 1.13

^a All EPCs were prepared using 20 mg ETP.

2.10.4. Treatment protocol

The mice were indiscriminately divided into 6 groups (seven mice each): Group 1 mice were administered inhalable spray-dried EPC/BER-loaded APBA-HSA mannitol nanocomposites (S4) eq. to 3 mg/kg of ETP (60 µg/mouse) using dry powder insufflator three times per week for two weeks. Group 2 mice were administered inhalable spray-dried physical mixture of free ETP/BER combination and mannitol at the same concentrations in nanocomposites by insufflation. Group 3 mice were injected in the tail vein with EPC/BER-loaded APBA-HSA NPs (T5) eq. to 3 mg/kg of ETP 3 times for two weeks. Group 4 mice were injected in the tail vein with combined free ETP/BER solution at the same concentrations in the NPs. Group 5; positive control (untreated chemically induced lung carcinoma bearing mice). Group 6; negative control (healthy mice).

For pulmonary administration, mice were placed in a supine position, to expose the trachea, after I.P. administration of ketamine (12.5 mg/kg) and xylazine (1.5 mg/kg) for anesthesia. Intratracheal administration was applied according to the method reported by Duret et al. [41]. For the formulations administered via the dry powder insufflations (DPI) method, an otoscope (Heine Mini 3000, Germany) was used to visualize tracheal openings. Pumping 0.2 ml of air contained in a syringe connected to the insufflator powder chamber was performed. Then, the powder chamber was weighed before and after powder filling and administration to detect the real quantity of sample released and aerosolized into the lungs.

2.10.5. In vivo anti-tumor efficacy

2.10.5.1. Lung and body weight. The mice body weight was measured periodically during the treatment. All the surviving mice were sacrificed after 21 days. The extracted lungs were isolated, weighed, morphologically examined and separated into two parts; one was for biological evaluation of tumor biomarkers after freezing at −80 °C and the other was for histopathological and immunohistochemical examination after fixation with 4% paraformaldehyde solution.

2.10.5.2. Tumor growth biomarkers. ELISA was used for the determination of the tumor growth biomarkers (Supplementary section).

2.10.5.3. Histopathological and immunohistochemical analysis. Histopathological changes, number of foci, their diameters and the proliferation extent indicated by Ki-67 protein were evaluated for the lung tumor samples (Supplementary section).

2.11. Statistics

Statistical analysis was done using GraphPad (Prism Version 5, CA, USA). Quantitative data were described using mean ± SD. For normally distributed quantitative data, comparisons among the different groups were done using analysis of variance (ANOVA; F test) with multiple comparisons between deposition data using a Tukey Kramer HSD test. Values for $P < .05$ were considered statistically significant.

3. Results and discussion

3.1. Preparation and characterization of EPC

As an approach to overcome its poor solubility, ETP-phospholipid complex (EPC) was formed via freeze-drying to obviate the heat used in solvent evaporation technique. For the preparation of EPC, dissolving and mixing the phospholipid and ETP together in a co-solvent was a prerequisite to prevent drug precipitation [32]. TBA was chosen as an organic co-solvent for lyophilization due to its high vapor pressure (41.25 mmHg at 25 °C), and high melting point (24 °C) [42]. In our preliminary trials, EPC was successfully prepared with 1:1, 1:2 and 1:4 molar ratios of ETP and phospholipid (Table 1). In contrast to 1:1 EPC (P1), the EPC prepared at the ratio of 1:2 and 1:4 (P2, P3) were very sticky and not easily reconstituted in water. This could be due to the excess unbound phospholipids that formed extra layers surrounding the complex as indicated by their high ζ potential values (−42.0 and −43.2 mV; respectively) (Fig. S1). Additionally, their PS and PDI were higher than P1, suggesting their non-homogeneity. Moreover, no difference was observed in their dissolution rate compared to P1 (Data not shown). Therefore, P2 and P3 were not recommended for formulation of aqueous HSA NPs.

Dissolution testing of the EPC in PBS (pH 7.4) showed improved drug dissolution via complexation with phospholipid (Fig. 1A). The enhanced dissolution of the drug complex can be explained by two factors. First, complexation of ETP with Lipoid S75 based on hydrogen bonding improved its hydrophilicity. Second, the reduced crystallinity of EPC contributed to the enhancement of ETP solubility (as indicated from DSC and XRD results). The presence of higher amount of mannitol obviously enhanced the dissolution rate of ETP. About 79.5% of ETP was dissolved after 2 h from 1:1 EPC (P4) prepared using 1% w/v mannitol compared to only 54.4% ETP dissolved from 1:1 EPC (P1) lyophilized using 0.5% w/v mannitol. The higher amount of mannitol, as a hydrophilic freeze-drying carrier, can increase the wettability and improve the dispersibility of ETP thus enhancing its dissolution profile [43]. The 1:1 EPC prepared using 1% mannitol (P4) exhibited a smaller particle size of 140 nm, ζ potential value of −27.0 mV and acceptable drug loading of 12.6% with enhanced dissolution and good powder flowability. Therefore; it was selected to be incorporated in HSA NPs (Fig. S2).

The FTIR spectrum of native ETP showed a broad characteristic band of phenolic –OH at 3412 cm^{−1}, an intense stretched band of ether in the 1660–1610 cm^{−1} region and C=O stretch band of ester bond at 1763 cm^{−1} (Fig. 1B) [44]. For Lipoid S75, peaks at the range from 1200 to 960 cm^{−1} were assigned to the phosphate group also shown in the physical mixture. The additive effect of both ETP and Lipoid S75 (3268.2 cm^{−1} of ETP-OH and 1762.21 cm^{−1} of the carbonyl in the acyl chains, respectively) was observed in the spectra of both physical mixture and EPC (P4). These findings suggested some weak physical interactions such as hydrogen bonding between the hydroxyl group of ETP and the phosphate group of phospholipid during complex formation [32].

The DSC thermogram of ETP showed a sharp endothermic melting peak at 275.13 °C (Fig. 1C). In the physical mixture thermogram, the peak of mannitol was still detectable at 166.076 °C but peaks'

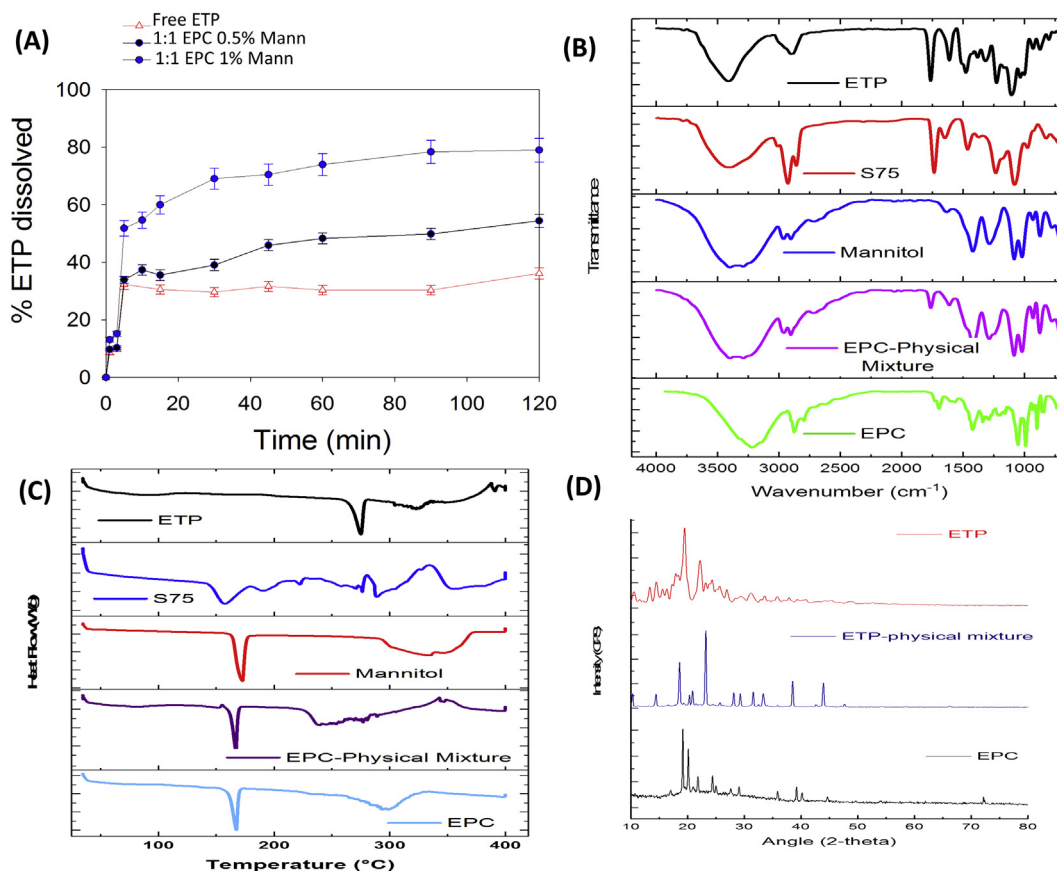


Fig. 1. Physicochemical properties of EPC; In vitro dissolution rate of ETP from the prepared EPC in PBS pH 7.4 at 37 °C (A), FTIR spectra of ETP, Lipid S75, mannitol, EPC and their physical mixture (B), their corresponding Differential scanning calorimetry (DSC) thermograms (C), their Powder X-ray diffractogram (D).

broadening was observed in the range of 230–330 °C may be attributed to the partial interaction between ETP and lipid. On the contrary, EPC thermogram showed the disappearance of the genuine peaks of ETP and lipid with a broad endothermal peak appearing at 297.77 °C. This could be explained by complex formation between ETP and lipid via hydrogen bonds and/or van der Waals forces [45]. As shown in XRD Fig. 1D, the crystalline state of ETP was confirmed by sharp peaks at $2\theta = 17.86^\circ$, 19.41° , and 22.17° . While the diffraction pattern of the physical mixture showed no modification of ETP crystallinity, XRD of EPC showed a relatively lower intensity peak of ETP; indicating reduction of its crystallinity upon formation of EPC.

3.2. Formulation and characterization HSA NPs

3.2.1. Particle size, DL, and EE

3.2.1.1. EPC-loaded HSA NPs. After pre-formulation of ETP as phospholipid complex (EPC, P4), it could be easily incorporated in HSA NPs via desolvation method. Desolvation process is well-controlled (pH, ethanol addition time, albumin concentration), robust process indicated by the little variability in PS, ZP, DL, and EE of drugs [46]. In our study, the NPs were obtained by the drop-wise addition of ethanol to continuously stirred aqueous solution of albumin containing EPC leading to the appearance of turbidity. EPC-loaded HSA NPs were then hardened using different crosslinkers (GA, GNP, and GA/Zn²⁺). The fabricated NPs (T1–T3) showed a size range of 136–190 nm with a highly negatively charged surface (−21.0 to −22.4 mV) indicating their high colloidal stability (Table 2). GA-crosslinked NPs (T1) were with 182.2 nm diameter and 70% EE of ETP. In comparison, HSA NPs crosslinked with GNP (T2) showed a smaller particle size (136 nm) but with a significantly lower EE (49.43%). In another strategy, the NPs were crosslinked with GA and further stabilized through Zn²⁺

coordinating bond. Zinc nitrate, as a bivalent metal ion, was reported to stabilize proteins to obtain preferable release behavior [47]. Consequently, HSA NPs (T3) with suitable PS of 190 nm were obtained but with a much lower drug EE (16%) may be due to the drug loss during the double centrifugation step. From all those results, GA-crosslinked NPs (T1) was selected as the optimized NPs due to their suitable size and high EE and DL.

3.2.1.2. EPC/BER co-loaded HSA NPs. To enhance the anti-tumor efficacy of ETP, the herbal drug BER was successfully co-loaded with EPC into HSA NPs using GA as a crosslinker. The EPC/BER co-loaded HSA NPs (T4) demonstrated a preferable PS of 173 nm and ζ potential of −24 mV with a high EE of both drugs (81.7 and 90.8% for BER and ETP, respectively). HSA offers a high capacity carrier for reversible binding to both drugs BER & ETP. BER is likely bound to the lipophilic cavity in HSA sub-domain IIA by the hydrophobic interaction in addition to the electrostatic force between the positively charged BER and the negatively charged HSA (at pH > pI = 4.9). Moreover, the amphiphilic property of phospholipid component of EPC not only facilitates its aqueous dispersion but also enables the hydrophobic binding to albumin binding domains.

To develop tumor-targeted NPs, APBA was coupled to HSA via simple carbodiimide reaction. The APBA-HSA conjugate was then used in the preparation of NPs incorporating BER/EPC (T5). The increase in particle size to 219 nm ensured the conjugation of the APBA onto the surface of HSA NPs. Conjugation of APBA on HSA was also confirmed by MALDI-TOF analysis revealing that the molecular mass of HSA and APBA-HSA were 68.83 and 71.04 kDa (m/z) respectively, thus suggesting that 16 molecules of APBA were conjugated to 1 HSA molecule (Fig. 2A). Conjugation of APBA to HSA was found to reduce the intrinsic tryptophan emission fluorescence intensity (Fig. 2B). This might be

Table 2

Composition and physicochemical properties of EPC-loaded & EPC/BER co-loaded HSA NPs.

	Cross-linker ^{a,b}	EE (%w/w)		DL (%w/w)		Particle size (nm)	PDI	Zeta potential (mV)
		BER	ETP	BER	ETP			
T1	GA (8%, 117 μ l)	–	70.00 $\pm$ 1.02	–	4.94 $\pm$ 0.13	182.2 $\pm$ 7.10	0.305 $\pm$ 0.09	–22.4 $\pm$ 0.89
T2	GNP (0.035 M)	–	49.43 $\pm$ 2.03	–	3.49 $\pm$ 0.22	136.0 $\pm$ 3.98	0.100 $\pm$ 0.02	–21.0 $\pm$ 1.16
T3	GA(8%, 117 μ l)/(0.025 M Zn)	–	16.00 $\pm$ 0.95	–	1.13 $\pm$ 0.10	190.0 $\pm$ 4.62	0.165 $\pm$ 0.02	–21.3 $\pm$ 1.03
T4	GA (8%, 117 μ l)	81.7 $\pm$ 2.64	90.80 $\pm$ 3.07	6.8 $\pm$ 0.21	5.85 $\pm$ 0.11	173.0 $\pm$ 2.50	0.132 $\pm$ 0.04	–24.0 $\pm$ 0.54
T5	GA (8%, 117 μ l)	60.5 $\pm$ 2.41	90.93 $\pm$ 2.87	5.2 $\pm$ 0.19	5.86 $\pm$ 0.10	219.0 $\pm$ 4.90	0.217 $\pm$ 0.08	–26.0 $\pm$ 1.42

^a All NPs were prepared with 200 mg HSA, 40 mg BER and 30 mg EPC except T5 was prepared by 200 mg APBA-HSA instead of HSA.^b T3 NPs were crosslinked with 117 μ l GA overnight and then separated by centrifugation. Then, the redispersed NPs were further stabilized by addition of 0.025 M zinc nitrate at ambient temperature under stirring for further 2 h then re-centrifuged to remove excess ions.

referred to the partial effect of APBA on the native packing of HSA without observation of denaturation (blue or red shift in the maximum λ_{em}) [48]. By the fact of the high affinity of boronic acid to Alizarin Red S dye (ARS), a change in the color of APBA-HSA conjugate from pink to slightly yellow accompanied with UV/VIS spectrum shift from 510 to 488 nm was recorded upon addition of ARS (Fig. S3 and S4) [49].

3.2.2. Morphological analysis

TEM micrographs revealed spherical shape and smooth surface non-targeted (T4) and APBA-HSA NPs (T5) with diameters of 148.3 and 201.21 nm, respectively which was quite smaller as compared with the results obtained from DLS technique (173 and 219 nm) (Fig. 3A and B). The hydrodynamic diameter of hydrated particles was measured by the laser scattering approach which was affected by the solvent phase [50]. In contrast, TEM measures the diameter of particles in the dried state. The conjugation of APBA onto the surface of HSA was also confirmed by the size increase of APBA-HSA NPs compared to the non-targeted HSA ones.

3.2.3. In vitro drug release

A sustained release behavior of ETP was observed from single-loaded EPC-HSA NPs (T1–T3), formulated with different crosslinkers over 3 days (Fig. 3C). GA was the most effective crosslinker with about 30.01 and 65.73% of ETP was released after 24 and 72 h, respectively without initial burst effect. First, ETP molecules were dissociated from EPC and distributed within the NPs, followed by the release of free ETP molecules from the NPs. Therefore, EPC acted as an additional physical

barrier to drug release [51]. In contrast, Wang et al. observed the biphasic release of ETP from BSA nanosuspensions prepared by high pressure homogenization with 2 h initial fast release of 25% followed by completely released drug after 24 h [52]. In another investigation, about 60% of ETP was released from PLGA NPs after 24 h and 100% was released after 72 h [53]. In the combined GA crosslinked BER/EPC HSA NPs (T4), a similar controlled release pattern of ETP up to 72 h was observed (Fig. 3D). On the other hand, a biphasic release of BER from HSA NPs (T4) was observed characterized by initial burst release of 41.9% during the first 3–4 h, followed by a second phase of slow release (45.3% of drug released after 72 h) (Fig. 3D). Notably, surface modification of HSA NPs by APBA (T5) did not much affect the release behavior of both drugs with the conservation of the sustained release pattern after APBA targeting.

Differences in the release pattern of the two drugs could be correlated to the differences in the physicochemical characteristics of the water-soluble BER (Log P $\sim$ –1.5) and the relatively hydrophobic ETP. The faster release of BER can sensitize tumor cells to ETP action by activation of ROS, besides its cytotoxic activity via topoisomerase inhibition thus retarding MDR. The delayed release of ETP could prevent premature drug release into circulation and maintain the synergistic efficacy at lower concentration and toxicity. A similar sequential release pattern was reported by Narayanan et al. where the faster release of epigallocatechin gallate from the shell of PLGA/casein core-shell NPs could sensitize tumor cells to the anti-cancer effect of the slowly released paclitaxel from its core [54].

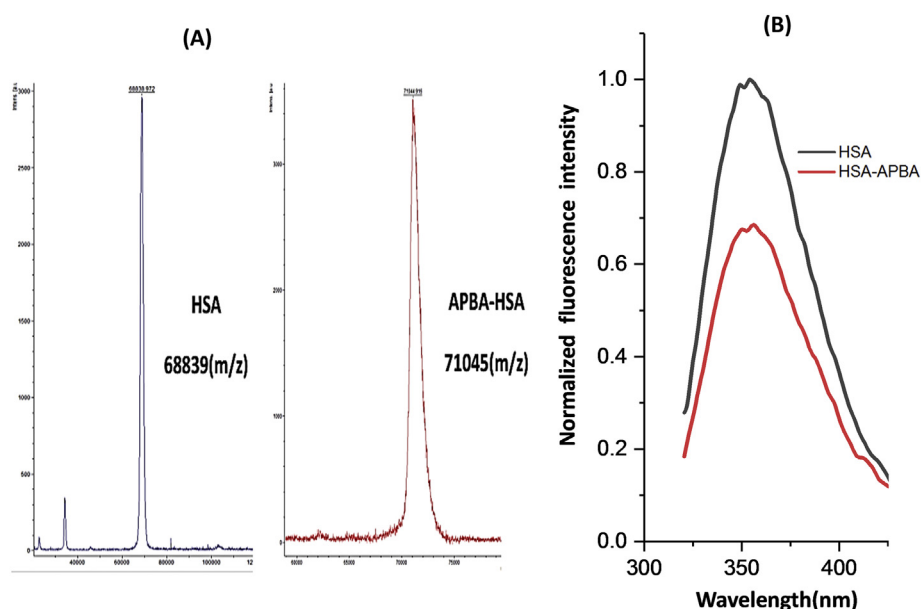


Fig. 2. Characterization of APBA-HSA conjugate; MALDI-TOF analysis of HSA and APBA-HSA (A). Intrinsic tryptophan emission fluorescence spectra of HSA and APBA-HSA were collected from 320 to 420 nm (λ_{ex} = 295) (B).

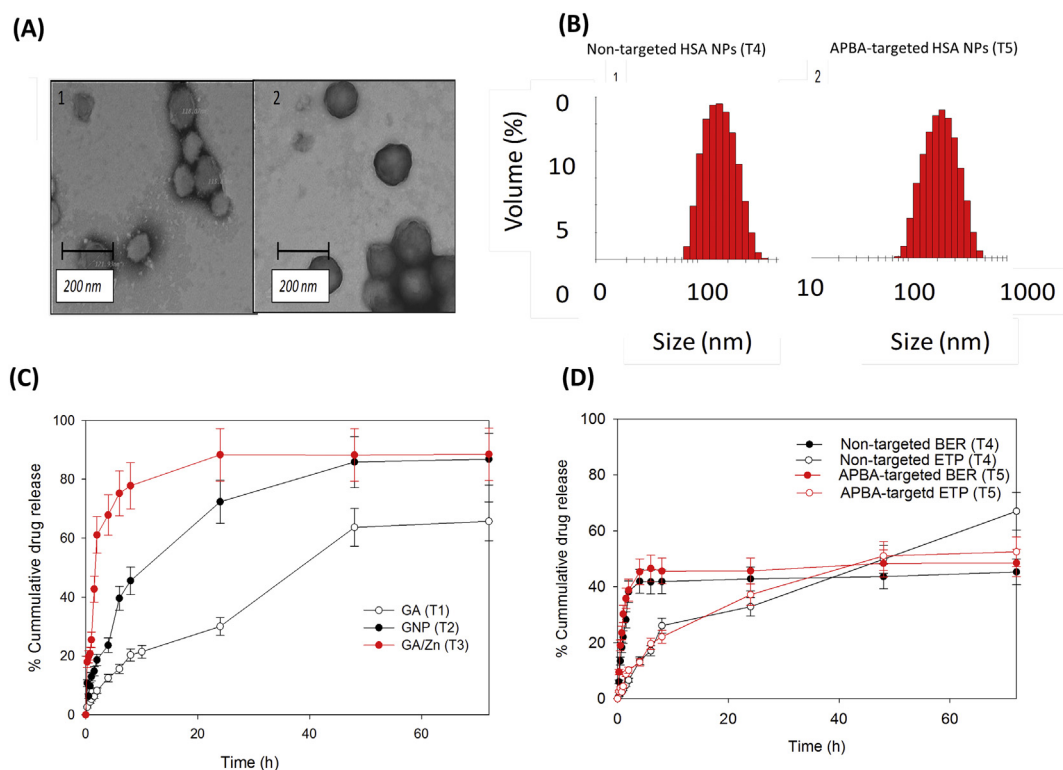


Fig. 3. Physicochemical properties of the prepared HSA NPs; TEM photographs (A) showing the morphology of HSA NPs (A1) non-targeted HSA NPs and (A2) APBA-targeted HSA NPs and their particle size distribution histograms (B). Release profiles of EPC from single-loaded EPC-HSA NPs, formulated with the different crosslinkers (C) and from the combined BER/ETP formulation (D) in PBS pH 7.4 at 37 °C.

3.3. *In vitro* cytotoxicity

Time- and concentration-dependency of the cytotoxic activity of ETP and BER, in both free and nano-encapsulated form, against A549 human lung cancer cells at 24 and 48 h was demonstrated (Fig. 4A and

B). The IC_{50} of non-targeted and APBA-targeted HSA NPs was 1.2- and 2-fold, respectively lower than free combined BER/ETP solution after 24 h (Fig. 4C). Compared to the NPs, the weak therapeutic activity of the free drugs could be correlated to the absence of controlled release characteristics and lack of targeting moieties, thus the cellular

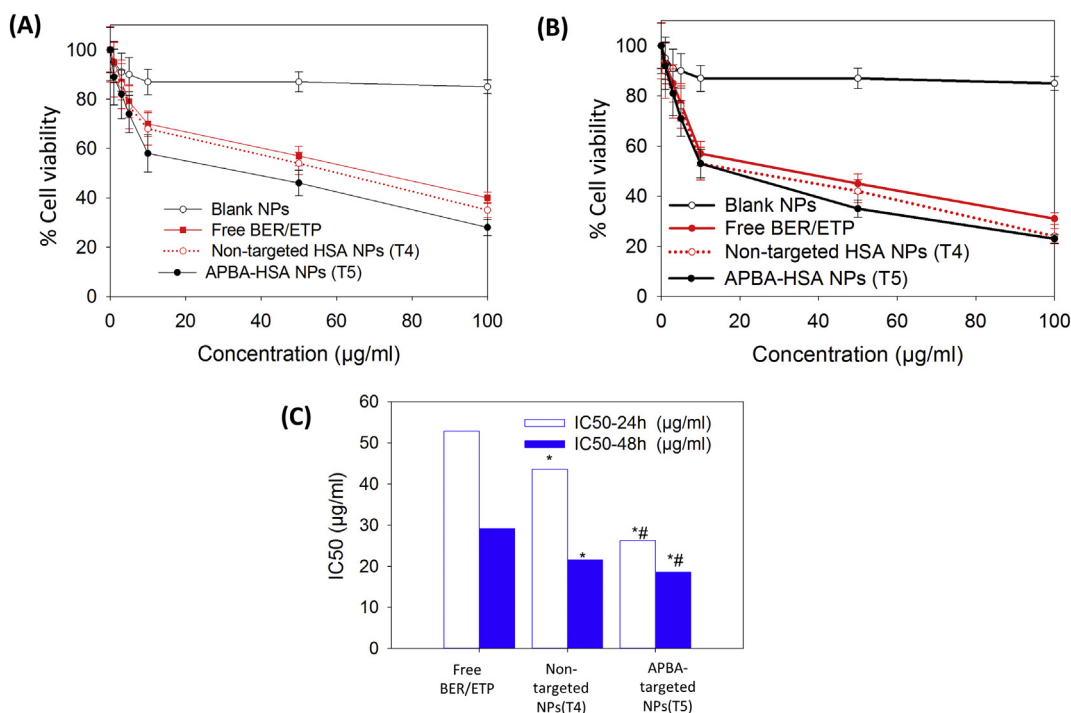


Fig. 4. Cytotoxicity analysis of free drugs and different BER/ETP HSA NPs on A549 human non-small cell lung cancer cell line at the concentration of 1–100 µg/ml after 24 h (A) and 48 h (B). IC_{50} (µg/ml) values of different BER/ETP-NCs HSA NPs (C). * $P < .05$ vs free ETP-BER, # $P < .05$ vs non-targeted HSA NPs.

internalization was only mediated by diffusion [55]. In contrast, the enhancement of the BER/ETP combination potency in NPs might be related to their efficient cellular internalization in tumor cells and the sustained drug release that preserves the drug level within cells. First, the intrinsic ability of HSA to bind with gp60 and SPARC on tumor cells contributed to the high cellular uptake [16]. Compared to non-targeted NPs, the inhibitory effect was increased by 1.6 times after anchoring APBA on the NPs surface. This amplified cytotoxicity of APBA-HSA NPs was attributed to sialic acid-dependent endocytosis by APBA, resulting in enhanced accumulation in tumor cells [26]. Moreover, complexation of ETP with phospholipid, the main constituent of the cell membrane, might enhance ETP permeability into the cells with increased cytotoxicity. A similar enhanced cytotoxic effect of camptothecin–phospholipid complex self-assembled NPs was reported [56].

After 48 h incubation with cells, the free combined ETP/BER solution showed more cytotoxic effect which ensured the time dependency of BER/ETP synergistic combination. The delayed release of ETP from NPs over 48 h enhanced its cytotoxic effect. The two formulations diminished the cell viability by the increased treatment time (Fig. 4B). These results were in agreement with a previously reported combination of ETP/quercetin loaded in PLGA NPs [11]. Safety of HSA NPs was confirmed by 85% viability of A549 human lung cancer cells after 48 h.

3.4. Qualitative analysis of cellular uptake

Confocal microscopy (CLSM) was used for tracking the cellular uptake of FITC-labeled HSA and APBA-HSA NPs into A549 lung cancer cells after incubation for 4 and 24 h. After 4 h, the DAPI stained nuclei (blue fluorescence) were surrounded by the green fluorescence of FITC-HSA and FITC-APBA-HSA NPs, indicating that sufficient quantity of the NPs could be internalized into the cytoplasm (Fig. 5). To exclude the role of albumin in cellular uptake by SPARC and gp60 and any interference by FITC (fluorescent probe), APBA-targeted NPs were compared to the non-targeted NPs and free FITC. Free FITC showed faded green fluorescence due to its relatively slow diffusion mechanism [39]. FITC-APBA-HSA NPs exhibited enhanced cellular uptake than FITC-HSA NPs, confirming the role of sialic acid-mediated endocytosis by APBA. Results showed that the cellular uptake was increased gradually in a time-dependent manner. After 24 h, the green fluorescence became denser

and well dispersed in the whole area of cytoplasm. Compared to the non-targeted NPs and free FITC, APBA-targeted NPs revealed elevated cellular internalization, which was indicated by intense green fluorescence. Thus, APBA-HSA NPs would offer a pathway to BER and EPC to enter and accumulate into the cells. This could explain the high cytotoxicity and low IC_{50} values when compared to free drugs.

3.5. Development and characteristics of inhalable APBA-HSA nanocomposites

Development of inhalable formulations provides a rational strategy for treatment of NSCLC. Localized pulmonary drug delivery could be utilized for decreasing the non-specific distribution of cytotoxic drugs, increasing its localization at the tumor site, and hence increasing the efficiency of treatment [1]. In the current study, dry powders embedding APBA-HSA NPs co-loaded with EPC and BER were developed via spray-drying as a speedy and single-step technique to form a dry powder of NPs embedded into microparticles (nanocomposites).

Yield and redispersibility: Carbohydrates are the most popular inhalation carriers, due to their ability to guard against drying. Upon inhalation, they dissolved rapidly releasing the incorporated NPs and preventing the phagocytic opsonization [57]. For albumin NPs, mannitol is more preferable than lactose, in order to avoid lactose interaction with albumin via Maillard reaction [58]. To target cancer cells, the nanocomposites have to recover their original nanosize after dissolving in the alveolar aqueous medium. Initial examination of mannitol as a pulmonary carrier at a fixed NPs/carrier ratio (1:5 w/w), with different total solid concentrations resulted in spray-dried nanocomposite powders with a yield range of 63.78–78.11% and good aqueous redispersibility ($RI = 1.19$ – 1.22) (Table 3, Fig. S5). Upon using the least aqueous volume for spray-drying to prevent the release of the water-soluble drug BER, a total solid concentration of 10.8% w/v was obtained. To further increase drug loading, mannitol was used at a lower NPs/carrier ratio (1:3 w/w), resulting in nanocomposites (F4) with the highest yield of 78.4% and superior aqueous redispersibility ($RI = 1.01$). Screening of different carbohydrate carriers suggested that mannitol is best recommended for our designed nanocomposites. Trehalose produced lower yield due to the cohesive nature of particles leading to powder adhesion to the spray-dryer wall while maltodextrin

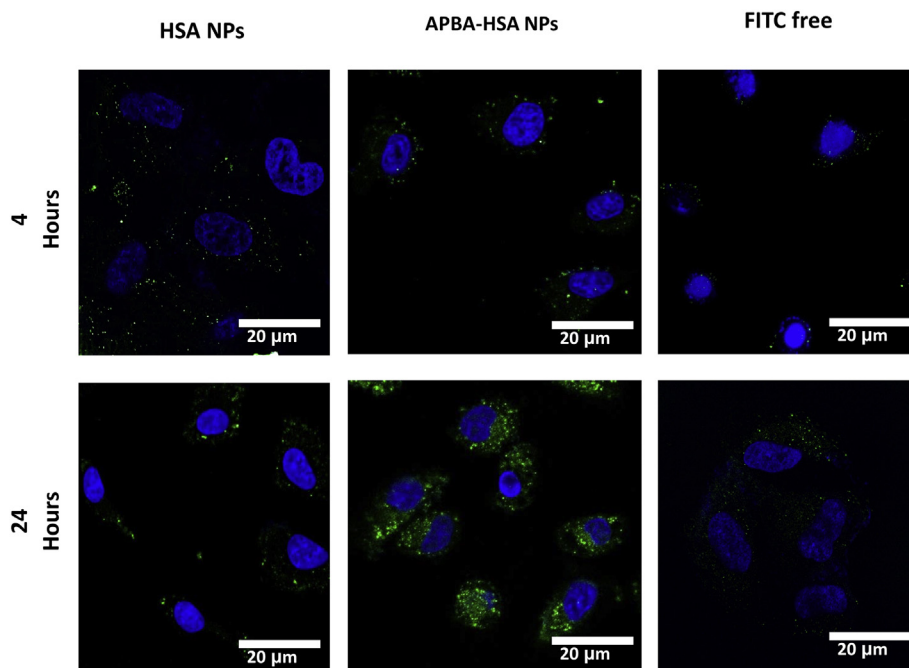


Fig. 5. Cellular uptake of non-targeted HSA NPs and APBA-HSA NPs with A549 cells after incubation for 4 h and 24 h.

Table 3

Composition, yield and aqueous re-dispersibility index of different prepared spray-dried nanocomposite formulations using different carriers.

Carrier ^a	Aqueous solution volume (ml)	Formula/carrier ratio (w/w)	Solid Conc. (w/v %)	Yield (%)	RI
F1 Mannitol	10.00	1:5	27.00	63.78	1.22
F2 Mannitol	25.00	1:5	10.80	78.00	1.18
F3 Mannitol	50.00	1:5	5.40	78.11	1.19
F4 Mannitol	25.00	1:3	7.20	78.40	1.01
F5 Maltodextrin	25.00	1:3	7.20	71.10	1.25
F6 Trehalose	25.00	1:3	7.20	60.00	1.20
F7 Leucine/Mannitol	25.00	1:1:2	7.20	70.40	1.50

^a RI was calculated as the ratio between the recovered PS of NPs from spray dried powder and the fresh NPs preparation.

gave lower yield and higher aqueous redispersibility, compared to mannitol.

Amino acids have been used, combined with polysaccharides, to reduce the adsorption of moisture and enhance the surface activity [59]. A carrier composed of leucine/mannitol in a ratio of 1:1:2 w/w (F7) was used to form nanocomposite powder with 70.4% yield and 1.5 RI. This relatively high RI value could be referred to the hydrophobic nature of leucine, which acted as a water repellent and reduced the sugars moisture uptake when re-suspended in aqueous medium [60]. Based on the investigation results of the effect of leucine addition to mannitol as a spray-drying carrier, F4 and F7 were selected.

SEM: The surface structure of spray-dried nanocomposites prepared using the two selected inhalation carriers: 1:3 w/w NPs/mannitol (F4) and 1:1:2 w/w NPs/leucine/mannitol (F7) was compared (Fig. 6). Addition of mannitol (F4) was found to produce spherical particles, with rough surfaces (Fig. 6A). The spherical wrinkled structure could be attributed to the

high vapor pressure, during spray-drying process, that leads to collapsing of the surface layer of particles [61]. On the other hand, the addition of leucine to mannitol (F7) produced hollow, collapsed doughnut-shaped particles (Fig. 6B). By virtue of leucine hydrophobicity and surface activity, it had a good propensity to be collected on the surface of the droplet; lessening their particle cohesiveness. The hydrodynamic impacts and the depletion of structural stability upon drying of the droplets lead to the formation of doughnut-like particles [62]. During fast solvent evaporation, the pressure obtained from the internal slow evaporation and gas expansion make the outer coat thinner, thus solvent diffused faster. Due to the impermeability of leucine, rupture of the particles occurred and the hollow structure was formed [63]. Compared to spherical particles, the high specific surface area of doughnut-shaped particles made it less desirable for nanoparticles encapsulation [64]. Producing rough surface particles contributes to the enhancement of the quality of dry powders for inhalation due to the reduced particles contact area and; thus decreases the particles adhesion [65]. Image analysis was performed, in order to compare the degree of surface roughness for the two selected formulations (F4 and F7). The 3D and 2D surface plots indicate that F7 has a relatively less corrugated surface as described by the green area in Fig. 6Bc and 6Cb.

Particle diameter: For efficient lung deposition, particles should have a volume median diameter value in the range of 1–5 μm . The size analysis data showed $D_{0.5}$ values of 2.5 and 2.82 μm for F4 and F7, respectively (Table 4); confirming their adequacy for pulmonary deposition. The slightly higher geometric diameter ($D_{0.5}$) for F7 may be due to the presence of leucine on the droplet surface during spray-drying that form the collapsed doughnut shape particles. For both F4 and F7, the particle size volume distribution was narrow and monomodal. Addition of leucine has reduced the value of span index from 1.58 (F7) to 1.98 (F4) indicating a better powder homogeneity [6]. Better inhalation performance in terms of higher pulmonary deposition was reported for formulations with lower span index [66].

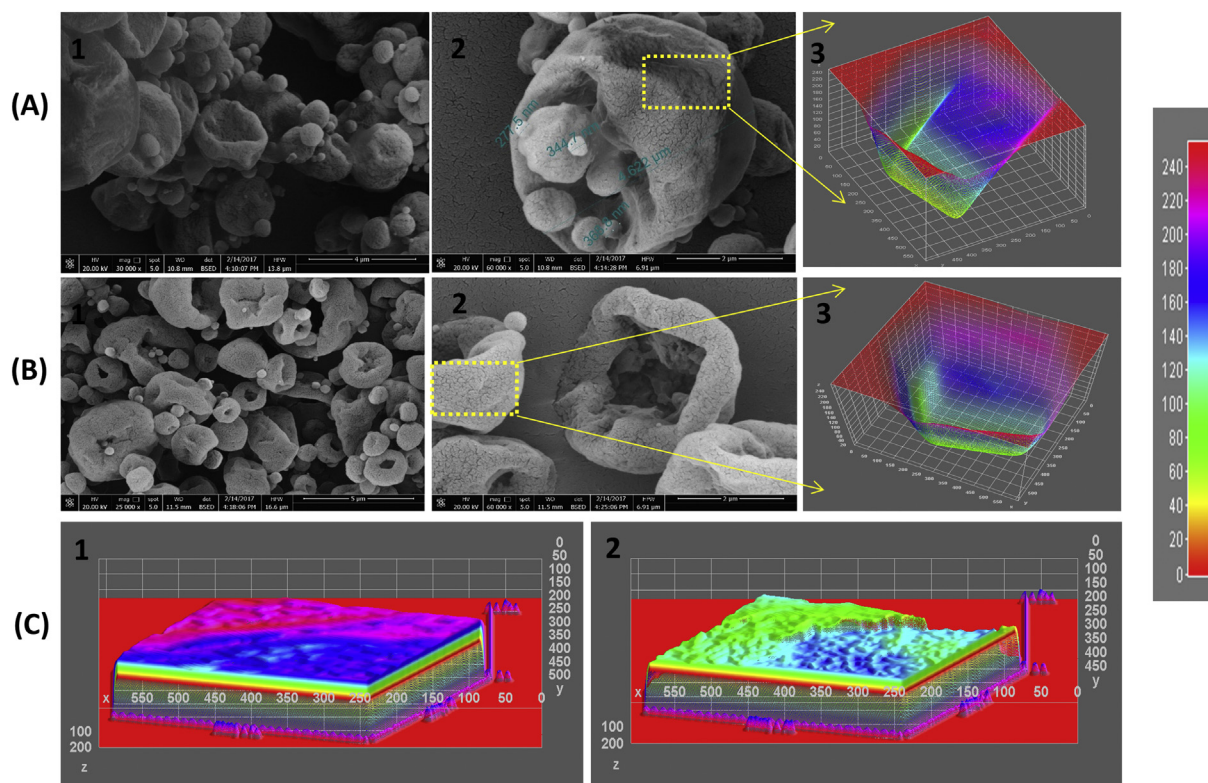


Fig. 6. Scanning electron micrographs of spray dried powder APBA-targeted HSA nanocomposites using mannitol as a carrier [(1:3), (A)] and APBA-targeted HSA nanocomposites using leucine and mannitol as a carrier [(1:1:2), (B)]. SEM Images Aa, Ba and Ab, Bb were captured at 30,000 $\times$ magnification and 60,000 $\times$ magnification, 2D plot (Ac, Bc) and 3D plot for mannitol carrier (Ca) and leucine and mannitol (Cb).

Table 4

Aerodynamic properties of spray-dried APBA-targeted HSA nanocomposites using mannitol (1:3 w/w) (F4) and leucine/mannitol (1:1:2 w/w) (F7) as pulmonary carriers.

	D _{0.5} (μm)	Span index	FPF (%)	ED (%)	MMAD (μm)	GSD
F4	2.50	1.99	77.86	80.45	2.12	2.13
F7	2.82	1.58	92.48	88.21	1.24	2.00

3.6. In vitro aerosol deposition

The effect of particle characteristics and carrier types on the in-vitro deposition of spray-dried HSA nanocomposites had been evaluated using ACI at a flow rate of 28.4 l/min (Fig. 7A). Results showed a high emitted dose percentage (ED = 80.45%) of the drugs from aerosolized powders containing mannitol (F4). The total amount of drugs deposited in stages 3–7 of ACI was considered as the FPF and was estimated to be 77.86%, with MMAD of 2.112 μm and GSD of 2.128 (Table 4). The surface properties of the particles related to mannitol carrier might be the reason for these high aerosolization parameters [67]. Leucine addition resulted in better aerosolization properties; indicated by the higher FPF value of 92.48% and ED of 88.21% with lower MMAD of 1.238 μm and GSD of 2.003 (Table 4). The enhanced aerosolization of nanocomposites prepared with leucine/mannitol combination (F7) could be referred to decreased moisture uptake due to the water-repellent action of leucine which might render the particles less cohesive [62]. In addition, leucine enhanced the dispersibility of the particles by shell formation and decreasing the surface free energy [68]. Additionally, the doughnut-shaped particles reduced inter-particulate contact surface areas, ensuring better particle dispersions [65].

Comparing with previously reported inhalable ETP delivery systems, inhalable freeze-dried ETP liposomes using various cryoprotectants at 1:10 ratio demonstrated particle diameter of 5.5 to 7.5 μm and FPF of about 50% [9]. Moreover, inhalable freeze-dried ETP-loaded PLGA microspheres, using 0.2% mannitol as a cryoprotectant, showed an average diameter of 11.9 μm indicating poor lung deposition [2]. On the contrary, our both nanocomposites powders (F4 and F7) exhibited appropriate characteristics for pulmonary drug delivery to lung cancer with acceptable diameter range (1–5 μm) and GSD range (1.3–3.0) for respirable particles.

Although nanocomposites prepared using leucine/mannitol (F7) showed better aerodynamic properties, it demonstrated high RI with

less effective dissolution of the complex carrier matrix which hindered the complete recovery of NPs to its original size (PS of the recovered NPs was 328.5 nm with high PDI). On the other hand, nanocomposites prepared using mannitol (F4) could recover the entrapped NPs with no change in their PS, with a rough surface and smaller calculated d_{aer} . Based on these results, the spray-dried APBA-HSA nanocomposites prepared using mannitol (F4) was selected as the optimized formulation for further in vivo investigations.

3.7. Drug release & long term stability

The effect of the spray-drying process on the release behavior of drugs was investigated in 37 °C in pH 7.4 PBS (Fig. 7B). Initially, about 40% of BER was burst released from spray-dried HSA nanocomposites (F4) during the first 6 h and then reached a plateau with about 46.7% of BER was released after 72 h. On the other hand, ETP showed sustained release with no initial burst effect with about 55.7% of the drug was released in 72 h, compared to 51.4% ETP released from HSA NPs (T5) before spray-drying. This confirms that the spray-dried nanocomposites maintained the sequential controlled release pattern of BER and ETP. After 4 months storage in a desiccator at room temperature, the spray-dried HSA nanocomposite powders appeared to maintain their original characteristics with no observed aggregation or clumps. After reconstitution in distilled water, a very slight increase in particle size to 225.2 nm with PDI of 0.173 was observed compared to the initial size of 219 nm. Interestingly, the spray-dried nanocomposites demonstrated a size of 234.8 nm with PDI of 0.205 and zeta potential of −23.6 mV after 1 year of storage thus confirming the long term stability of spray-dried HSA nanocomposites. The solid state characterization confirmed the absence of any drug-carrier interactions (discussed in details in the Supplementary section).

3.8. In vivo lung deposition and retention study

Cascade impactors do not adequately simulate the real in vivo conditions and give no idea about the retaining efficiency of the particles after pulmonary administration. So, in vivo testing should be performed to confirm particle retention in lung tissues [69]. The fluorescent microscopic images of mice lung sections at different time intervals following powder insufflation of spray-dried FITC-labeled APBA-HSA nanocomposite powder and free FITC powder are shown in Fig. 8A. Maximum fluorescence intensity was observed after 4 h of pulmonary administration of FITC-labeled nanocomposites, indicating

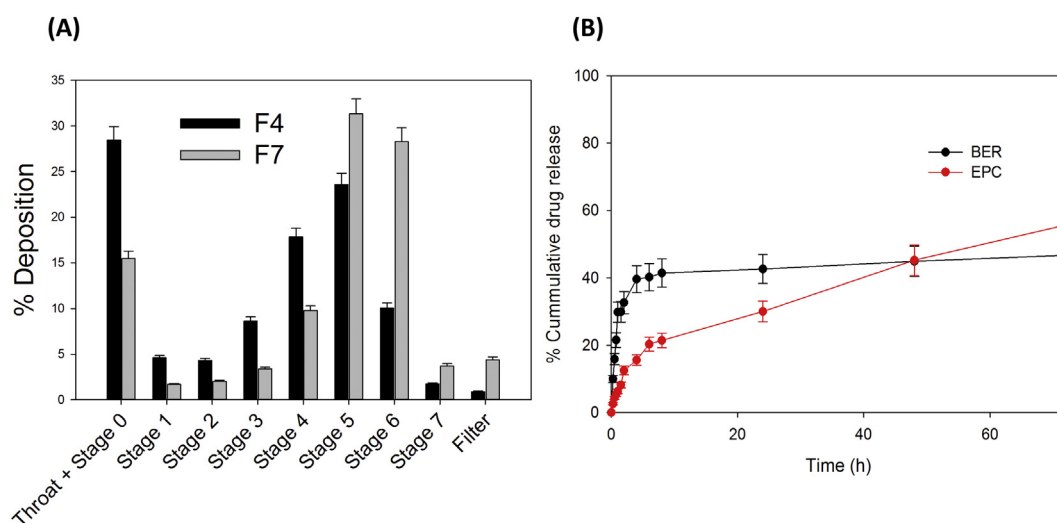


Fig. 7. In vitro aerosol deposition profiles of spray dried APBA-targeted HSA nanocomposites (A) using mannitol as a carrier [(1:3), F4] and APBA-targeted HSA nanocomposites using leucine and mannitol as a carrier [(1:1:2), F7] using Andersen Cascade Impactor (ACI). Experiments were performed at air flow rate 28.5 l/min. The release profile of spray dried albumin nanoparticles loaded BER/EPC using mannitol as a carrier with (1:3) weight ratio in PBS pH 7.4 at 37 °C (B).

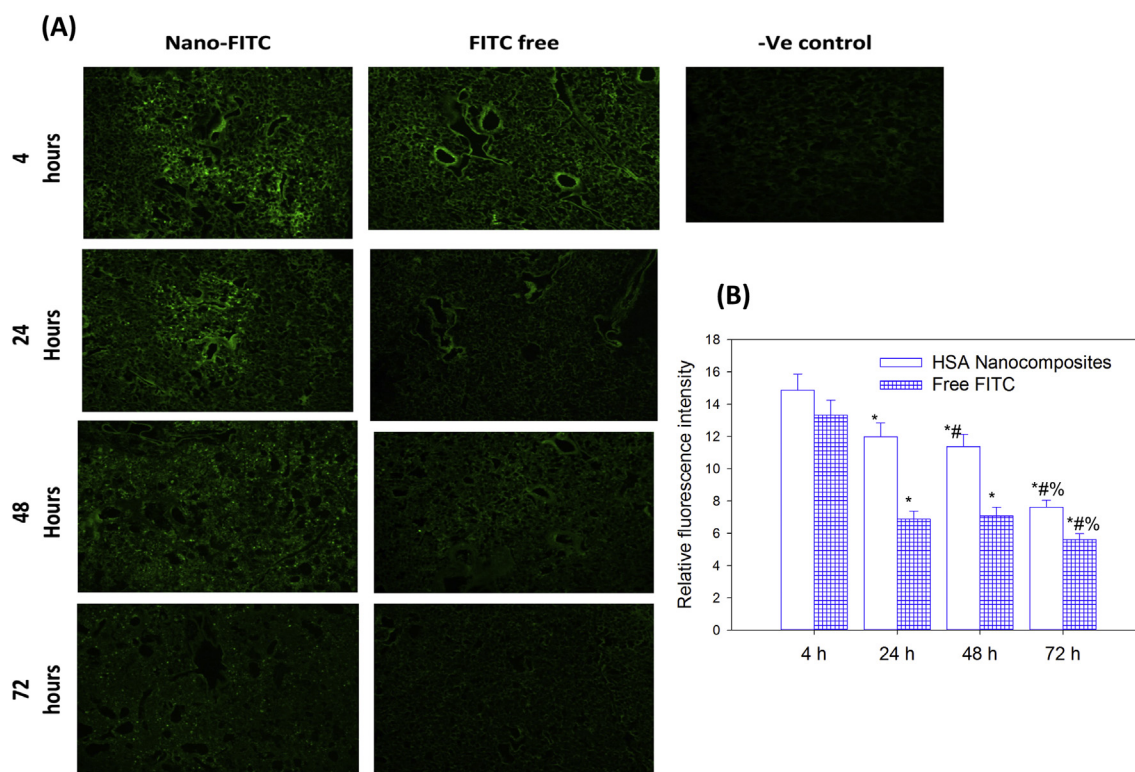


Fig. 8. Fluorescent images of cross-section of lung at 4, 24, 48, 72 h after intratracheal administration of APBA-targeted HSA nanocomposites powder labeled by FITC dye (original magnification, 10) versus free FITC dye (A), and their relative fluorescence intensity vs time (B). * $P < .05$ vs 4 h, # $P < .05$ vs 24 h, % $P < .05$ vs 48 h.

their quickly distribution and deposition into the lungs (Fig. 8A and B). Free FITC powder exhibited lower inhalation efficiency when compared to nanocomposite powder reflecting the in vivo lung deposition superiority of the nanocomposites. Autofluorescence of the native lung tissues (the control group) was neglected (autocorrected). By time, the fluorescence intensity and particles number in lung sections were declined in both FITC-APBA-HSA nanocomposites and free FITC powder, suggesting their biodegradation and clearance from lung tissues. Although the fluorescence was faded after 72 h but nanocomposites were still detectable with higher intensity than free FITC (Fig. 8B). These results ensured the appropriate aerodynamic parameters of spray-dried nanocomposite powder for in vivo pulmonary delivery of ETP and BER to the deep lung tissues and being well-retained over 3 days, confirming the efficiency of treatment.

3.9. In vivo anti-tumor efficacy

3.9.1. Safety and therapeutic activity evaluation by determination of body and lung weight

The wet lung weight was determined by weighing the isolated lungs after removal of the trachea and other extraneous tissue to eliminate any interference (Fig. 9A and B). Untreated positive control mice displayed the highest lung weight compared to the different treatment groups, indicating lung damage due to tumor growth, excluding any formulation inflammatory effect. There was no significant difference in the weight of lungs between the insufflated HSA nanocomposites (F4) and normal negative control group ($p < .05$), with 2.8-fold reduction in lung weight ($p < .05$) compared to the positive control group. This confirmed the nanocomposite ability of suppressing the tumor growth in lungs. Mice treated with inhalable free drugs powder and I.V. HSA NPs (T5) showed 1.89- and 2.03-fold reduction in lung weight ($p < .05$) compared to the positive control group. Notably, cancerous lesions in huge number were observed in positive control extracted lungs, whereas native control lungs showed normal physiological

features. Compared to the positive control and I.V. HSA NPs, inhalable nanocomposites powder displayed superior treatment strategy with visibly much reduced tumor tissues number (Fig. 9B).

Mice body weight was monitored during the treatment (Fig. 9C). There was approximately no change in the body weights of mice treated with the inhalable nanocomposites powder at the end of 21 days, confirming the safety of the inhalable formulations. Group treated with inhalable free drugs powder showed 31.5% reduction of body weight during the treatment while the positive control group exhibited only 13.2% reduction, indicating the possible absorption of drugs into systemic circulation. In comparison with the inhalable nanocomposites, I.V. HSA NPs administered with the same dose showed more reduction in the body weight, suggesting the superiority of the inhalable nanocomposite powder formulation.

3.9.2. Effect of different treatments on lung homogenate VEGF and caspase

Relatively to the positive control group, the inhalable nanocomposite powder treatment group showed a 3.64-fold reduction in the VEGF expression level, compared to only 1.87- fold reduction in I.V. HSA NPs group ($P < .05$) confirming a potential reduction effect on tumor angiogenesis by the inhalation route of administration over the I.V. route. Inhalable free drugs powder could only exhibit a 1.6-fold reduction of VEGF level which ensured their inadequacy in inhibition of tumor growth (Fig. 9D).

As shown in Fig. 9E, all treated groups exhibited a higher apoptotic effect compared to untreated positive control group. An enhancement in apoptosis activation ($p < .05$) was observed in mice treated with inhalable nanocomposites powder (16.8 ng/g tissue) over the inhalable free drugs powder treated group (13.08 ng/g tissue) and the positive group (1.66 ng/g tissue). In addition, injectable HSA NPs treatment succeeded to significantly ($p < .05$) exhibited higher Casp-3 levels compared to I.V. free drugs and control group (Fig. 9E).

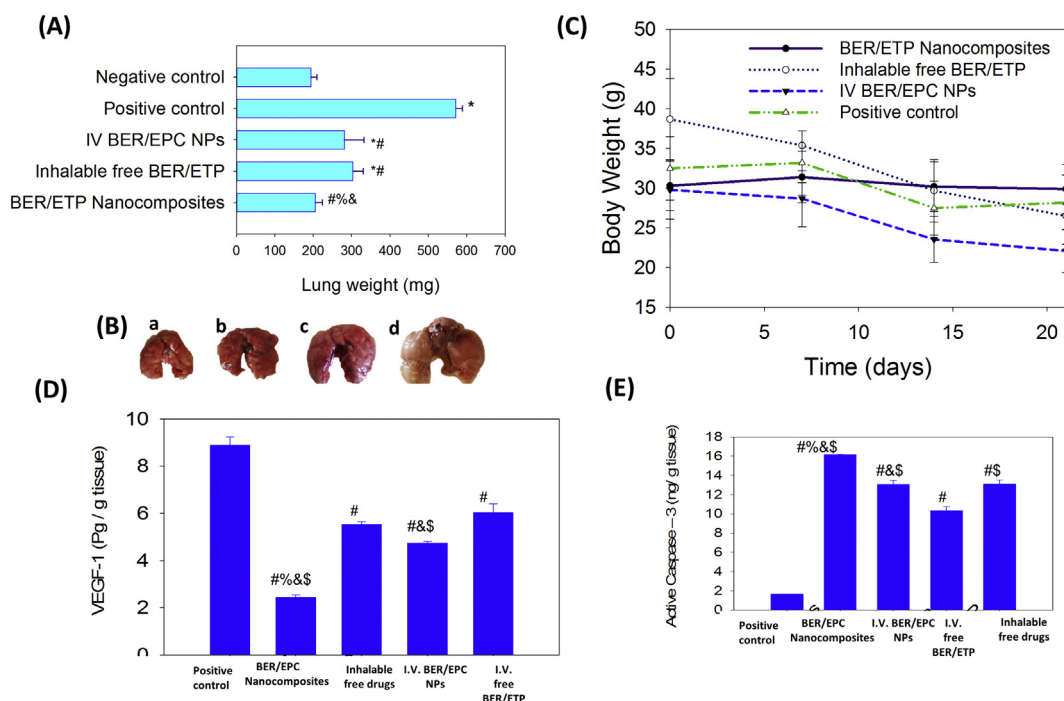


Fig. 9. Lung weights of the different groups post 3 weeks treatment (I.V./inhalation) compared to untreated positive control (A), Excised lungs from different treatment groups (B); negative control (Ba), inhalable APBA-targeted HSA nanocomposites (Bb) I.V. APBA-targeted HSA NPs (Bc) and positive control (Bd). Body weights measurements of mice at indicated time-points along the treatment duration with different treatments (I.V./inhalation) compared to untreated positive control (C), and the level of the tumor markers including Caspase-3 (D), and VEGF (E) in urethane induced lung cancer bearing mice treated with different treatments compared to untreated positive control. * $P < .05$ vs negative control, # $P < .05$ vs positive control, & $P < .05$ vs inhalable free drugs, % $P < .05$ vs I.V. APBA-targeted HSA NPs, \$ $P < .05$ vs I.V. free drugs.

3.9.3. Histopathological and immunohistochemical analysis

As shown in Fig. 10A, normal epithelial single-layered alveolar tissue was observed in the negative control group, while the positive control group showed epithelial hyperplasia to adenoma (blue arrow), hemorrhage (white arrow) and inflammatory cells infiltration (black arrow) as signs of pre-cancerous and cancerous tissues upon histopathological examination. Compared to I.V. HSA NPs, the inhalable nanocomposites-treated group revealed superlative enhancement in the histopathological neoplastic insult with reduced signs of cancerous transformation. Additionally, inhalable nanocomposites and I.V. NPs treated groups exhibited a significant 12.25- and 2.58-fold reduction in the number of metastatic lung foci, respectively with highly reduced diameter (0.62 and 2.85 mm, respectively), compared to untreated control group ($P < .05$), confirming the efficacy of the inhalation route of administration in treatment of the lung cancer (Fig. 10B).

Additionally, immunohistochemical examination of the proliferation marker Ki-67 confirmed the superiority of inhalable nanocomposites, over other tested routes of administration (I.V.) and the control group, indicated by the reduction in brown staining density (Fig. 10A). Significant reduction in Ki-67-expression was detected in inhalable nanocomposites-treated mice (16.8%) than I.V. NPs and untreated control group (52.6, and 92.94%, respectively) ($P < .05$) (Fig. 10C).

From all these results, it was confirmed that dry nanocomposite powder inhalational therapy was superior in anti-tumor efficacy, anti-angiogenic, apoptotic and anti-proliferative activity than the I.V. NPs and both inhalable and injectable free drugs. The substantial improvement in suppressing tumor growth is likely due to the local, concentrated and deep lung deposition treatment achieved with inhalable nanocomposites that maintained targeted effective drug levels over long periods to kill tumor cells.

4. Conclusion

To the best of our knowledge, there is no inhalable ETP delivery system for local delivery to the deep lungs was evaluated in lung cancer bearing mice. In the present study, we proposed inhalable APBA-targeted HSA nanocomposites for combined delivery of BER and ETP to urethane-induced lung cancer bearing mice. First, ETP was incorporated within albumin nanoparticles via its pre-formulation as phospholipid complex (EPC). Second, BER and EPC were successfully co-encapsulated into albumin nanoparticles with high drug loading and suitable size. The tumor-targeting effect of APBA could enhance the internalization of the nanocarriers and hence their cytotoxicity against lung cancer cells. Finally, the APBA-HSA nanoparticles were spray-dried into inhalable nanocomposites for deep lung deposition. The spray-dried nanocomposites prepared using mannitol carrier displayed favorable aerosolization performance with small MMAD (2.11 μ m) and high FPF (77.86%) as well as high aqueous redispersibility. The superior anti-tumor efficacy of the inhalable nanocomposites was revealed by reduction of lung weights and growth biomarkers in lung cancer bearing mice and augmented by histopathological and immunohistochemical analysis. The inhalable dose of ETP administered to mice was much lower than that used in other studies via systemic routes. Overall, the developed inhalable dry powder APBA-targeted HSA nanocomposites co-loaded with ETP and BER could present a promising alternative to systemic therapy of lung cancer. However, further studies should be performed in the future on large animal models that may have significantly different characteristics and profiles from small animals (particles deposition, internalization, and risk/cost-benefit ratio). Long-term efficacy and safety including inflammation and immunogenicity could be evaluated specifically in repeated-cycles treatment.

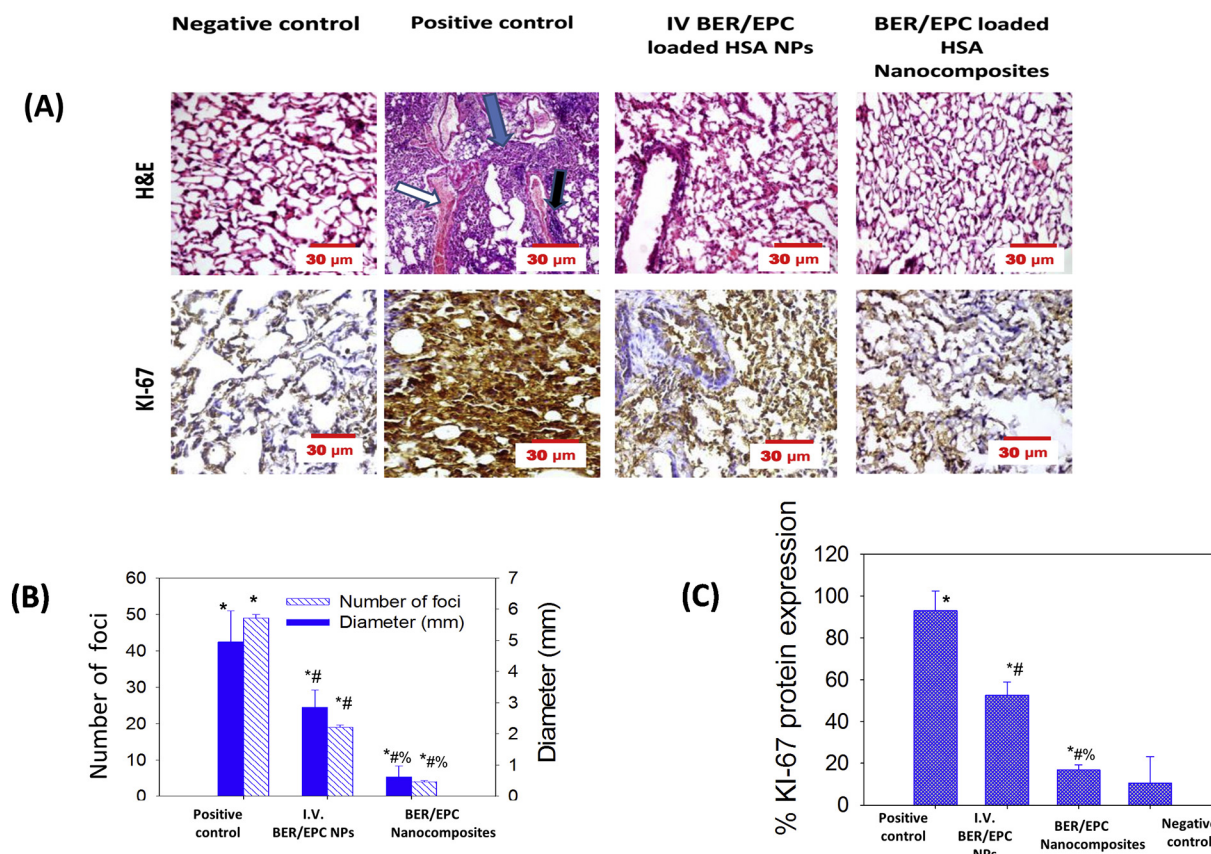


Fig. 10. Histopathological and Immunohistochemical analysis of lung sections of I.V. APBA-targeted HSA NPs and inhalable APBA-targeted HSA nanocomposites treatment groups compared to untreated positive control and healthy negative control groups; H&E and Ki-67 expression (A), Number of foci and their diameter related to H&E sections (B) and % expression of proliferating protein Ki-67 related to Ki-67 sections (C). * $P < .05$ vs negative control, # $P < .05$ vs positive control, % $P < .05$ vs I.V. APBA-targeted HSA NPs.

Acknowledgement

This work was supported by the research grant (No. 5731) of Science and Technology Development Fund (STDF), Ministry of Scientific Research, Egypt.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jconrel.2018.07.014>.

References

- H.M. Abdelaziz, M. Gaber, M.M. Abd-Elwakil, M.T. Mabrouk, M.M. Elgohary, N.M. Kamel, D.M. Kabary, M.S. Freag, M.W. Samaha, S.M. Mortada, K.A. Elkhodairy, J.Y. Fang, A.O. Elzoghby, Inhalable particulate drug delivery systems for lung cancer therapy: nanoparticles, microparticles, nanocomposites and nanoaggregates, *J. Control. Release* 269 (2017) 374–392.
- R. Feng, Z. Zhang, Z. Li, G. Huang, Preparation and in vitro evaluation of etoposide-loaded PLGA microspheres for pulmonary drug delivery, *Drug Deliv.* 21 (2014) 185–192.
- S. Sabra, M. Abdelmoneem, M. Abdelwakil, M. Mabrouk, D. Anwar, R. Mohamed, S. Khattab, A. Bekhit, K. Elkhodairy, M. Freag, A.O. Elzoghby, Self-assembled nanocarriers based on amphiphilic natural polymers for anti-cancer drug delivery applications, *Curr. Pharm. Des.* 23 (2017) 5213–5229.
- S.N. Khattab, S.E. Abdel Naim, M. El-Sayed, A.A. El Bardan, A.O. Elzoghby, A.A. Bekhit, A. El-Faham, Design and synthesis of new s-triazine polymers and their application as nanoparticulate drug delivery systems, *New J. Chem.* 40 (2016) 9565–9578.
- A.O. Elzoghby, Editorial: polymeric nanocarriers as robust platforms for cancer therapy, *Curr. Pharm. Des.* 23 (2017) 5211–5212.
- T. Peng, S. Lin, B. Niu, X. Wang, Y. Huang, X. Zhang, G. Li, X. Pan, C. Wu, Influence of physical properties of carrier on the performance of dry powder inhalers, *Acta Pharm. Sin. B* 6 (2016) 308–318.
- C.L. Tseng, S.Y. Wu, W.H. Wang, C.L. Peng, F.H. Lin, C.C. Lin, T.H. Young,

M.J. Shieh, Targeting efficiency and biodistribution of biotinylated-EGF-conjugated gelatin nanoparticles administered via aerosol delivery in nude mice with lung cancer, *Biomaterials* 29 (2008) 3014–3022.

- Y. Abbas, H.M. Azzazy, S. Tammam, A. Lamprecht, M.E. Ali, A. Schmidt, S. Sollazzo, S. Mathur, Development of an inhalable, stimuli-responsive particulate system for delivery to deep lung tissue, *Colloids Surf. B: Biointerfaces* 146 (2016) 19–30.
- J. Parmar, D.J. Singh, A.A. Lohade, D.D. Hegde, P.S. Soni, A. Samad, M.D. Menon, Inhalational system for etoposide liposomes: formulation development and in vitro deposition, *Indian J. Pharm. Sci.* 73 (2011) 656–662.
- Z. Heger, H. Polanska, S. Krizkova, J. Balvan, M. Raudenska, S. Dostalova, A. Moullick, M. Masarik, V. Adam, Co-delivery of VP-16 and Bcl-2-targeted antisense on PEG-grafted mWCNTs for synergistic in vitro anti-cancer effects in non-small and small cell lung cancer, *Colloids Surf. B: Biointerfaces* 150 (2017) 131–140.
- S. Pimple, A.S. Manjappa, M. Ukawala, R.S. Murthy, PLGA nanoparticles loaded with etoposide and quercetin dihydrate individually: in vitro cell line study to ensure advantage of combination therapy, *Cancer Nanotechnol.* 3 (2012) 25–36.
- U. Schroeder, Hydrolytically activated etoposide prodrugs inhibit MDR-1 function and eradicate established MDR-1 multidrug-resistant T-cell leukemia, *Blood* 102 (2003) 246–253.
- L. Gianni, E. Munzone, G. Capri, F. Fulfaro, E. Tarenzi, F. Villani, C. Spreafico, A. Laffranchi, A. Caraceni, C. Martini, Paclitaxel by 3-hour infusion in combination with bolus doxorubicin in women with untreated metastatic breast cancer: high antitumor efficacy and cardiac effects in a dose-finding and sequence-finding study, *J. Clin. Oncol.* 13 (1995) 2688–2699.
- C. He, R. Rong, J. Liu, J. Wan, K. Zhou, J.X. Kang, Effects of Coptis extract combined with chemotherapeutic agents on ROS production, multidrug resistance, and cell growth in A549 human lung cancer cells, *Chin. Med.* 7 (2012) 11.
- B.S. Anisha, D. Sankar, A. Mohandas, K.P. Chennazhi, S.V. Nair, R. Jayakumar, Chitosan-hyaluronan/nano chondroitin sulfate ternary composite sponges for medical use, *Carbohydr. Polym.* 92 (2013) 1470–1476.
- A.O. Elzoghby, M.M. Elgohary, N.M. Kamel, Implications of protein- and peptide-based nanoparticles as potential vehicles for anticancer drugs, *Adv. Protein Chem. Struct. Biol.* 98 (2015) 169–221.
- A. Elzoghby, Editorial (thematic issue: nanocarriers based on natural polymers as platforms for drug and gene delivery applications), *Curr. Pharm. Des.* 22 (2016) 3303–3304.
- M. Socinski, Update on taxanes in the first-line treatment of advanced non-small-cell lung cancer, *Curr. Oncol.* 21 (2014) e691–e703.

- [19] K.R. Vinod, S. Sandhya, J. Chandrashekar, R. Swetha, T. Rajeshwar, B. David, S. Anbuazaghan, Review on genesis and characterization of phytosomes, *ILPSRR* 4 (2010) 69–75.
- [20] A.O.M. Elzoghby, S. Khamees, M.W. Helmy, M.A. ElDemellawy, S.A. Sheweita, Superiority of aromatase inhibitor and cyclooxygenase-2 inhibitor combined delivery: hyaluronate-targeted versus PEGylated protamine nanocapsules for breast cancer therapy, *Int. J. Pharm.* 529 (2017) 178–192.
- [21] S.W. El-Far, M.W. Helmy, S.N. Khatib, A.A. Bekhit, A.A. Hussein, A.O. Elzoghby, Phytosomal bilayer-enveloped casein micelles for codelivery of monascus yellow pigments and resveratrol to breast cancer, *Nanomedicine (London)* 13 (2018) 481–499.
- [22] X. He, N. Xiang, J. Zhang, J. Zhou, Y. Fu, T. Gong, Z. Zhang, Encapsulation of teniposide into albumin nanoparticles with greatly lowered toxicity and enhanced antitumor activity, *Int. J. Pharm.* 487 (2015) 250–259.
- [23] D.M. Kabary, M.W. Helmy, K.A. Elkhodairy, J.Y. Fang, A.O. Elzoghby, Hyaluronate/lactoferrin layer-by-layer-coated lipid nanocarriers for targeted co-delivery of rapamycin and berberine to lung carcinoma, *Colloids Surf. B: Biointerfaces* 169 (2018) 183–194.
- [24] S.A. Sabra, A.O. Elzoghby, S.A. Sheweita, M. Haroun, M.W. Helmy, M.A. ElDemellawy, Y. Xia, D. Goodale, A.L. Allan, S. Rohani, Self-assembled amphiphilic zein-lactoferrin micelles for tumor targeted co-delivery of rapamycin and wogonin to breast cancer, *Eur. J. Pharm. Biopharm.* 128 (2018) 156–169.
- [25] J.Y. Lee, S.J. Chung, H.J. Cho, D.D. Kim, Phenylboronic acid-decorated chondroitin sulfate A-based theranostic nanoparticles for enhanced tumor targeting and penetration, *Adv. Funct. Mater.* 25 (2015) 3705–3717.
- [26] X. Wang, H. Tang, C. Wang, J. Zhang, W. Wu, X. Jiang, Phenylboronic acid-mediated tumor targeting of chitosan nanoparticles, *Theranostics* 6 (2016) 1378–1392.
- [27] H. Jasmina, O. Džana, E. Alisa, V. Edina, R. Ognjenka, Preparation of nanoemulsions by high-energy and lowenergy emulsification methods, in: A. Badnjec (Ed.), *CMBEBIH 2017*, Springer Nature Singapore Pte Ltd., 2017, pp. 317–322.
- [28] F. Donsi, B. Senatore, Q. Huang, G. Ferrari, Development of novel pea protein-based nanoemulsions for delivery of nutraceuticals, *J. Agric. Food Chem.* 58 (2010) 10653–10660.
- [29] Y. Wei, Y. Wang, D. Xia, S. Guo, F. Wang, X. Zhang, Y. Gan, Thermosensitive liposomal codelivery of HSA-paclitaxel and HSA-Ellagic acid complexes for enhanced drug perfusion and efficacy against pancreatic cancer, *ACS Appl. Mater. Interfaces* 9 (2017) 25138–25151.
- [30] A. Varki, R. Schauer, Sialic acids, In: A. Varki, R.D. Cummings, J.D. Esko, H.H. Freeze, P. Stanley, C.R. Bertozzi, G.W. Hart, M.E. Etzler Editors. *Essentials of Glycobiology* 2nd Ed. Cold Spring Harbor (NY): Cold Spring Harbor Laboratory Press, (2009) (Chapter 14).
- [31] J.C. Sung, B.L. Pulliam, D.A. Edwards, Nanoparticles for drug delivery to the lungs, *Trends Biotechnol.* 25 (2007) 563–570.
- [32] M.S. Freag, Y.S. Elnaggar, O.Y. Abdallah, Lyophilized phytosomal nanocarriers as platforms for enhanced diosmin delivery: optimization and ex vivo permeation, *Int. J. Nanomedicine* 8 (2013) 2385–2397.
- [33] A.O. Elzoghby, S.A. El-Lakany, M.W. Helmy, M.M. Abu-Serie, N.A. Elgindy, Shell-crosslinked Zein nanocapsules for oral codelivery of exemestane and resveratrol in breast cancer therapy, *Nanomedicine (London)* 12 (2017) 2785–2805.
- [34] S. Dreis, F. Rothweiler, M. Michaelis Jr., J. Cinatl, J. Kreuter, K. Langer, Preparation, characterisation and maintenance of drug efficacy of doxorubicin-loaded human serum albumin (HSA) nanoparticles, *Int. J. Pharm.* 14 (2007) 341 (207).
- [35] A. Jahanban-Esfahlan, S. Dastmalchi, S. Davaran, A simple improved desolvation method for the rapid preparation of albumin nanoparticles, *Int. J. Biol. Macromol.* 91 (2016) 703–709.
- [36] J. Liu, B. Zhang, Z. Luo, X. Ding, J. Li, L. Dai, J. Zhou, X. Zhao, J. Ye, K. Cai, Enzyme responsive mesoporous silica nanoparticles for targeted tumor therapy in vitro and in vivo, *Nanoscale* 7 (2015) 3614–3626.
- [37] A.O.M. Elzoghby, S. Khamees, M.W. Helmy, M.A. ElDemellawy, S.A. Sheweita, Multi-reservoir phospholipid shell encapsulating protamine nanocapsules for co-delivery of letrozole and celecoxib in breast cancer therapy, *Pharm. Res.* 34 (2017) 1956–1969.
- [38] N. Alam, R.D. Dubey, A. Kumar, M. Koul, N. Sharma, P.R. Sharma, B.K. Chandan, S.K. Singh, G. Singh, P.N. Gupta, Reduced toxicological manifestations of cisplatin following encapsulation in folate grafted albumin nanoparticles, *Life Sci.* 142 (2015) 76–85.
- [39] R.D. Dubey, N. Alam, A. Saneja, V. Khare, A. Kumar, S. Vaidh, G. Mahajan, P.R. Sharma, S.K. Singh, D.M. Mondhe, P.N. Gupta, Development and evaluation of folate functionalized albumin nanoparticles for targeted delivery of gemcitabine, *Int. J. Pharm.* 492 (2015) 80–91.
- [40] M.A. Hamzawy, A.M. Abo-Youssef, H.F. Salem, S.M. Ali, An additional risk of lung cancer from recurrent exposure to ethyl carbamate (EC) in BALB/C mice, *J. Cancer Sci. Therap.* (2015) (2015) 7–11.
- [41] C. Duret, N. Wauthoz, R. Merlos, J. Goole, C. Maris, I. Roland, T. Sebti, F. Vanderbist, K. Amighi, In vitro and in vivo evaluation of a dry powder endotracheal insufflator device for use in dose-dependent preclinical studies in mice, *Eur. J. Pharm. Biopharm.* 81 (2012) 627–634.
- [42] D.L. Teagarden, D.S. Baker, Practical aspects of lyophilization using non-aqueous co-solvent systems, *Eur. J. Pharm. Sci.* 15 (2002) 115–133.
- [43] A. Das, A.K. Nayak, B. Mohanty, S. Panda, Solubility and dissolution enhancement of etoricoxib by solid dispersion technique using sugar carriers, *ISRN Pharm.* 2011 (2011) 819765.
- [44] W.Y. Qian, D.M. Sun, R.R. Zhu, X.L. Du, H. Liu, S.L. Wang, pH-sensitive strontium carbonate nanoparticles as new anticancer vehicles for controlled etoposide release, *Int. J. Nanomedicine* 7 (2012) 5781–5792.
- [45] H. Chen, S. Shi, M. Zhao, L. Zhang, H. He, X. Tang, A lyophilized etoposide sub-micron emulsion with a high drug loading for intravenous injection: preparation, evaluation, and pharmacokinetics in rats, *Drug Dev. Ind. Pharm.* 36 (2010) 1444–1453.
- [46] H. Kouchakzadeh, M.S. Safavi, S.A. Shojaosadati, Efficient delivery of therapeutic agents by using targeted albumin nanoparticles, *Adv. Protein Chem. Struct. Biol.* 98 (2015) 121–143.
- [47] C. Li, L. Xing, S. Che, Coordination bonding based pH-responsive albumin nanoparticles for anticancer drug delivery, *Dalton Trans.* 41 (2012) 3714–3719.
- [48] A.I. Ledesma-Osuna, G. Ramos-Clamont, L. Vazquez-Moreno, Biorecognition of chemically modified bovine serum albumin with lactose prepared under different conditions, *J. Agric. Food Chem.* 57 (2009) 9734–9739.
- [49] S.N. Narla, P. Pinnamaneni, H. Nie, Y. Li, X.L. Sun, BSA-boronic acid conjugate as lectin mimetics, *Biochem. Biophys. Res. Commun.* 443 (2014) 562–567.
- [50] W. She, K. Luo, C. Zhang, G. Wang, Y. Geng, L. Li, B. He, Z. Gu, The potential of self-assembled, pH-responsive nanoparticles of mPEGylated peptide dendron-doxorubicin conjugates for cancer therapy, *Biomaterials* 34 (2013) 1613–1623.
- [51] J. Xie, Y. Li, L. Song, Z. Pan, S. Ye, Z. Hou, Design of a novel curcumin-soybean phosphatidylcholine complex-based targeted drug delivery systems, *Drug Deliv.* 24 (2017) 707–719.
- [52] Z. Wang, Z. Li, D. Zhang, L. Miao, G. Huang, Development of etoposide-loaded bovine serum albumin nanosuspensions for parenteral delivery, *Drug Deliv.* 22 (2014) 79–85.
- [53] K.S. Yadav, K.K. Sawant, Formulation optimization of etoposide loaded PLGA nanoparticles by double factorial design and their evaluation, *Curr. Drug Deliv.* 7 (2010) 51–64.
- [54] S. Narayanan, M. Pavithran, A. Viswanath, D. Narayanan, C.C. Mohan, K. Manzoor, D. Menon, Sequentially releasing dual-drug-loaded PLGA-casein core/shell nanomedicine: design, synthesis, biocompatibility and pharmacokinetics, *Acta Biomater.* 10 (2014) 2112–2124.
- [55] Y.C. Kuo, Y.C. Chen, Targeting delivery of etoposide to inhibit the growth of human glioblastoma multiforme using lactoferrin- and folic acid-grafted poly(lactide-co-glycolide) nanoparticles, *Int. J. Pharm.* 479 (2015) 138–149.
- [56] Z. Fan, G. Liu, Y. Li, J. Ma, J. Lin, F. Guo, Z. Hou, L. Xie, Self-assembly of the active lactone form of a camptothecin-phospholipid complex for sustained nuclear drug delivery, *RSC Adv.* 6 (2016) 82949–82960.
- [57] H. Zhang, C.P. Hollis, Q. Zhang, T. Li, Preparation and antitumor study of camptothecin nanocrystals, *Int. J. Pharm.* 415 (2011) 293–300.
- [58] C. Sinsuebpol, J. Chatchawalsaisin, P. Kulvanich, Preparation and in vivo absorption evaluation of spray dried powders containing salmon calcitonin loaded chitosan nanoparticles for pulmonary delivery, *Drug Des. Devel. Ther.* 7 (2013) 861–873.
- [59] T. Rattanupatam, T. Srichana, Budesonide dry powder for inhalation: effects of leucine and mannitol on the efficiency of delivery, *Drug Deliv.* 21 (2014) 397–405.
- [60] Y. Wang, K. Kho, W.S. Cheow, K.A. Hadinoto, Comparison between spray drying and spray freeze drying for dry powder inhaler formulation of drug-loaded lipid-polymer hybrid nanoparticles, *Int. J. Pharm.* 424 (2012) 98–106.
- [61] R. Osman, K.T. Al Jamal, P.L. Kan, G. Awad, N. Mortada, A.E. El-Shamy, O. Alpar, Inhalable DNase I microparticles engineered with biologically active excipients, *Pulm. Pharmacol. Ther.* 26 (2013) 700–709.
- [62] S.H. Lee, D. Heng, W.K. Ng, H.K. Chan, R.B. Tan, Nano spray drying: a novel method for preparing protein nanoparticles for protein therapy, *Int. J. Pharm.* 403 (2011) 192–200.
- [63] K. Kanogwan, T. Srichana, Nano spray-dried pyrazinamide-l-leucine dry powders, physical properties and feasibility used as dry powder aerosols, *Pharm. Dev. Technol.* 21 (2016) 68–75.
- [64] B.D. Nandiyanto, K. Okuyama, Progress in developing spraydrying methods for the production of controlled morphology particles: from the nanometer to sub-micrometer size ranges, *Int. J. Antimicrob. Agents* 22 (2011) 1–19.
- [65] N.Y. Chew, P. Tang, H.K. Chan, J.A. Raper, How much particle surface corrugation is sufficient to improve aerosol performance of powders? *Pharm. Res.* 22 (2005) 148–152.
- [66] N.Y. Chew, H.K. Chan, Effect of powder polydispersity on aerosol generation, *J. Pharm. Pharm. Sci.* 5 (2002) 162–168.
- [67] N.Y. Chew, H.K. Chan, Use of solid corrugated particles to enhance powder aerosol performance, *Pharm. Res.* 18 (2001) 1570–1577.
- [68] K. Kho, K. Hadinoto, Optimizing aerosolization efficiency of dry-powder aggregates of thermally-sensitive polymeric nanoparticles produced by spray-freeze-drying, *Powder Technol.* 214 (2011) 169–176.
- [69] K. Nahar, N. Gupta, R. Gauvin, S. Absar, B. Patel, V. Gupta, A. Khademhosseini, F. Ahsan, In vitro, in vivo and ex vivo models for studying particle deposition and drug absorption of inhaled pharmaceuticals, *Eur. J. Pharm. Sci.* 49 (2013) 805–818.